

The Gauss-Manin connection via the de Rham space

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ABSTRACT. We give a simple, geometric description of the Gauss-Manin connection in terms of the *de Rham space* X_{dR} .

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1. INTRODUCTION

Let k be a field of characteristic 0 and let $f: X \rightarrow S$ be a morphism of smooth k -schemes. It is classical result that the relative algebraic de Rham cohomology $H_{\mathrm{dR}}^i(X/S)$ carries a natural flat connection, the *Gauss–Manin* connection.

Analytically, we can see this as follows: suppose $k = \mathbb{C}$ and let $f^h: \mathfrak{X} \rightarrow \mathfrak{S}$ be the associated holomorphic map between smooth proper complex analytic varieties. Then, we have an isomorphism

$$H_{\mathrm{dR}}^i(\mathfrak{X}/\mathfrak{S}) \cong R^i f_*^h \mathbb{C}_{\mathfrak{X}} \otimes_{\mathbb{C}_{\mathfrak{S}}} \mathcal{O}_{\mathfrak{S}}$$

Assuming f^h is a proper submersion, $R^i f_*^h \mathbb{C}_{\mathfrak{X}}$ is a locally constant sheaf by the Ehresmann fibration theorem. Then there is an analytic flat connection

$$(\mathrm{id} \otimes d): H_{\mathrm{dR}}^i(\mathfrak{X}/\mathfrak{S}) \rightarrow H_{\mathrm{dR}}^i(\mathfrak{X}/\mathfrak{S}) \otimes \Omega_{\mathfrak{S}}^1$$

by the Riemann-Hilbert correspondence, which in turn gives a flat connection on $H_{\mathrm{dR}}^i(X/S)$ by GAGA.

The algebraicity of the Gauss-Manin connection is more subtle. Katz and Oda provide one such proof of this by the observation that the connection appears as a differential d_1^{0i} on the E_1 page of a certain spectral sequence [KO68]. The aim of this note is to provide another such construction, this time coming from Simpson’s

de Rham space X_{dR} . In particular, we explain how the de Rham cohomology of X/S , considered as a $\mathcal{D}_S$ -module under the Gauss-Manin connection, is isomorphic to the higher direct image of $\mathcal{O}_{X_{\text{dR}}}$ along the induced morphism between de Rham spaces $f_{\text{dR}} : X_{\text{dR}} \rightarrow S_{\text{dR}}$.

2. THE DE RHAM SPACE

2.1. Preliminaries. We introduce the de Rham space, originally defined by Simpson [Sim96]. Many of the technical arguments in this section are found in [Rib24].

Let k be a fixed field of characteristic 0 and Aff_k be the category of affine k -schemes. Let τ be a Grothendieck topology on Aff_k . We define a τ algebraic stack to be a sheaf $\text{Aff}_k \rightarrow \text{Grpd}$ in the τ topology. The category of τ algebraic stacks will be denoted AlgStk_τ .

Definition 2.1 (de Rham space). Given any presheaf $X : \text{Aff}_k \rightarrow \text{Set}$, we define X_{dR} to be the presheaf

$$X_{\text{dR}}(A) := X(A^{\text{red}})$$

Note that there is a natural map $X \rightarrow X_{\text{dR}}$ and that associated to a morphism $X \rightarrow S$ there is an induced morphism $X_{\text{dR}} \rightarrow S_{\text{dR}}$. In this case, we define the relative de Rham space $(X/S)_{\text{dR}}$ to be the fibered product $X_{\text{dR}} \times_{S_{\text{dR}}} S$.

Let X be a scheme which is formally smooth over k . If a k -algebra A has nilpotent nilradical (for instance, if A is Noetherian), then every A^{red} -point of X extends to an A -point,

$$\begin{array}{ccc} \text{Spec } A^{\text{red}} & \xrightarrow{\quad} & X \\ \downarrow & \nearrow \text{dashed} & \downarrow \\ \text{Spec } A & \xrightarrow{\quad} & \text{Spec } k \end{array}$$

by an infinitesimal lifting argument. Consequently, the natural map $X \rightarrow X_{\text{dR}}$ is an epimorphism of presheaves. In general though, it suffices to have X/k smooth.

Proposition 2.2. *If X is smooth over k , then $X \rightarrow X_{\text{dR}}$ is an epimorphism.*

Proof. Let $\mathfrak{J}$ be the poset of nilpotent ideals $I \subset A$. Then $A^{\text{red}} \cong \varinjlim_{I \in \mathfrak{J}} A/I$, and

$$\text{Hom}(\text{Spec } A^{\text{red}}, X) \cong \text{Hom}(\varinjlim_{I \in \mathfrak{J}} \text{Spec } A/I, X) \cong \varinjlim_{I \in \mathfrak{J}} \text{Hom}(\text{Spec } A/I, X)$$

where the second isomorphism holds because X is locally of finite presentation [Sta18] Tag 01ZC. Consider an element on the right; this is given by a representative $\text{Spec } A/I \rightarrow X$ for some nilpotent ideal I . This lifts to $\text{Spec } A \rightarrow X$ by the formal smoothness of X . $\square$

From the above proof, we also have the following characterization: if X is locally of finite presentation over k , then

$$X_{\text{dR}}(A) \cong \text{Hom}(\varinjlim_{I \in \mathfrak{J}} \text{Spec } A/I, X) \cong \varinjlim_{I \in \mathfrak{J}} \text{Hom}(\text{Spec } A/I, X)$$

and therefore X_{dR} is a colimit. An important consequence of this is the following.

Corollary 2.3. *If X is locally of finite presentation over k , $X_{\mathrm{dR}} : \mathrm{Aff}_k \rightarrow \mathrm{Set}$ commutes with finite limits and arbitrary colimits.*

An important property of the de Rham space is that it satisfies descent.

Proposition 2.4 ([Rib24], Proposition 1.1.14). *If X is a locally finite type scheme, the de Rham space X_{dR} satisfies fppf descent.*

We emphasize that k must have characteristic 0. Thus by our conventions, X_{dR} is an fppf algebraic stack.

2.2. Algebraic de Rham cohomology. A key feature of the de Rham space is that the cover $X \rightarrow X_{\mathrm{dR}}$ provides an identification of the algebraic de Rham cohomology of X with the $\mathcal{O}$ -coherent cohomology of X_{dR} . We will prove a relative version of this theorem.

Let $f : X \rightarrow S$ be a morphism of k -schemes, and consider the induced morphisms

$$\begin{array}{ccc} (X/S)_{\mathrm{dR}} & \longrightarrow & X_{\mathrm{dR}} \\ \tilde{f} \downarrow & & \downarrow f_{\mathrm{dR}} \\ S & \longrightarrow & S_{\mathrm{dR}} \end{array}$$

Let $\mathcal{O}_{\mathrm{dR}/S}$ be the structure sheaf of $(X/S)_{\mathrm{dR}}$.

Theorem 2.5. *If $f : X \rightarrow S$ is a morphism of smooth k -schemes, then*

$$H_{\mathrm{dR}}^i(X/S) \cong R^i \tilde{f}_* \mathcal{O}_{\mathrm{dR}/S}$$

where the left is the Zariski hypercohomology of $\Omega_{X/S}^\bullet$.

To establish 2.5, we need the following (both of which are proved in [Rib24]).

Lemma 2.6. *Let X be locally of finite presentation. Then, the formal completion of X along a closed subscheme Z may be identified as*

$$\widehat{X}_Z \cong X \times_{X_{\mathrm{dR}}} Z_{\mathrm{dR}}$$

Lemma 2.7. *If X and S are locally of finite presentation,*

$$X \times_{(X/S)_{\mathrm{dR}}} X \cong (\widehat{X \times_S X})_\Delta$$

where the right-hand side is the formal completion of the diagonal $X \rightarrow X \times_S X$.

Some remarks are in order. A priori, the definition of formal completion of X along a subscheme Z requires Z to be closed. However in 2.6, note that the right-hand side does not require $Z \rightarrow X$ to be a closed immersion. Thus in 2.7, we do not require $X \rightarrow S$ to be separated and instead define formal completion as in 2.6.

Proof of 2.7. Let $Y = X \times_S X$, so that $\widehat{Y}_\Delta \cong Y \times_{Y_{\text{dR}}} X_{\text{dR}}$. The result follows from

$$\begin{array}{ccc} X \times_{(X/S)_{\text{dR}}} X & \longrightarrow & Y = X \times_S X \\ \downarrow & & \downarrow \\ X_{\text{dR}} & \xrightarrow{\Delta} & Y_{\text{dR}} \cong X_{\text{dR}} \times_{S_{\text{dR}}} X_{\text{dR}} \end{array}$$

being cartesian. $\square$

Proof of 2.5. For simplicity, take $S = \text{Spec } k$. Since $\widehat{X \times X} \rightrightarrows X \rightarrow X_{\text{dR}}$ is a coequalizer of presheaves, $X \rightarrow X_{\text{dR}}$ is an effective epimorphism. Therefore the simplicial presheaf $\widehat{X}^\bullet = \check{\text{Cech}}(X \rightarrow X_{\text{dR}})$ is a hypercover of X_{dR} . This gives

$$H^i(X_{\text{dR}}, \mathcal{O}_{X_{\text{dR}}}) \cong H^i(\widehat{X}^\bullet, \mathcal{O}_{\widehat{X}^\bullet})$$

By 2.7, $\widehat{X}^n$ is the ind-scheme given by formal completion of the n -fold product $X \times \cdots \times X$ along the diagonal. This simplicial scheme is used by Grothendieck to calculate the cohomology of the *infinitesimal site* in [Gro68]. This hinges on the observation that $\widehat{X}^\bullet$ gives a hypercover of the terminal object in the infinitesimal topos. Precisely, there is a spectral sequence

$$E_2^{ij} = H^i(n \mapsto H^j(\widehat{X}^{n+1}, \mathcal{O}_{\widehat{X}^{n+1}})) \implies H^{i+j}(X_{\text{inf}}, \mathcal{O}_{X_{\text{inf}}})$$

which degenerates and yields an canonical isomorphism

$$H^i(X_{\text{inf}}, \mathcal{O}_{X_{\text{inf}}}) \cong H^i(n \mapsto \mathcal{O}(\widehat{X}^{n+1})) = H^i(\widehat{X}^\bullet, \mathcal{O}_{\widehat{X}^\bullet})$$

because $H^j(\widehat{X}^{n+1}, \mathcal{O}_{\widehat{X}^{n+1}}) = 0$ for $j > 0$. It's a deeper result that for smooth schemes in characteristic 0, there is a canonical isomorphism

$$H^i(X_{\text{inf}}, \mathcal{O}_{X_{\text{inf}}}) \cong H_{\text{dR}}^i(X/k)$$

(Theorem 4.1 in [Gro68]). The exact same proof works over a general base S . $\square$

2.3. Quasi-coherent sheaves on de Rham spaces. A key insight of Simpson was that for a smooth k -schemes X , that the category of quasi-coherent sheaves on X_{dR} identifies with the category of algebraic left $\mathcal{D}_X$ -modules on X . Since

$$\widehat{X \times X} \rightrightarrows X \rightarrow X_{\text{dR}}$$

is a coequalizer of presheaves, we may consider X_{dR} as a quotient of X by the relation of *infinitesimal closeness*. The data of a quasi-coherent sheaf on X_{dR} is thus equivalent to a quasi-coherent sheaf on X along with a compatible system of isomorphisms over points which lie in some n -fold formal thickening of the diagonal. This is equivalent to an action of the ring of n -th order differential operators on X .

Theorem 2.8. *If X is a smooth k -scheme, there is an equivalence*

$$\Phi : \mathrm{QCoh}(X_{\mathrm{dR}}) \rightarrow \mathcal{D}_X\text{-mod}$$

between the category of quasi-coherent sheaves on X_{dR} and the category of quasi-coherent left $\mathcal{D}_X$ -modules.

This equivalence can be upgraded to a *cohomological* equivalence. Given a left $\mathcal{D}_X$ -module $(\mathcal{F}, \nabla)$ on X , define the de Rham complex $\mathrm{dR}_X(\mathcal{F})$

$$\mathcal{F} \rightarrow \mathcal{F} \otimes_{\mathcal{O}_X} \Omega_X^1 \rightarrow \mathcal{F} \otimes_{\mathcal{O}_X} \Omega_X^2 \rightarrow \cdots$$

by iterating ∇ . Then the hypercohomology of this complex is isomorphic to the cohomology of the associated quasi-coherent sheaf on X_{dR} .

Proposition 2.9. *If X is a smooth k -scheme and $(\mathcal{F}, \nabla)$ is a left $\mathcal{D}_X$ -module,*

$$\mathbb{H}^i(X, \mathrm{dR}_X(\mathcal{F})) \cong H^i(X_{\mathrm{dR}}, \Phi(\mathcal{F}, \nabla))$$

Remark 2.10. Since $\Phi(\mathcal{O}_{X_{\mathrm{dR}}}) \cong (\mathcal{O}_X, d : \mathcal{O}_X \rightarrow \Omega_X^1)$, this recovers 2.5.

Let S be a smooth k -scheme. Then 2.8 gives an equivalence of functors $F \cong \pi^*$ where F is the forgetful functor $\mathcal{D}_S\text{-mod} \rightarrow \mathrm{QCoh}(S)$ and π is the natural morphism $S \rightarrow S_{\mathrm{dR}}$. Therefore, any quasi-coherent sheaf $\mathcal{F}$ on S can be endowed with a flat connection precisely when it is in the essential image of π^* . Then the fact that the relative de Rham cohomology sheaf $H_{\mathrm{dR}}^i(X/S)$ has a flat connection must correspond to the fact that it is pulled back from some quasi-coherent sheaf on S_{dR} .

Consider the diagram,

$$\begin{array}{ccc} (X/S)_{\mathrm{dR}} & \xrightarrow{\tilde{\pi}} & X_{\mathrm{dR}} \\ \tilde{f} \downarrow & & \downarrow f_{\mathrm{dR}} \\ S & \xrightarrow{\pi} & S_{\mathrm{dR}} \end{array} \quad (*)$$

and recall we have a natural map

$$\pi^* R^i f_{\mathrm{dR}*} \mathcal{O}_{X_{\mathrm{dR}}} \rightarrow R^i \tilde{f}_* \tilde{\pi}^* \mathcal{O}_{X_{\mathrm{dR}}} \cong H_{\mathrm{dR}}^i(X/S)$$

called the base-change morphism.

Proposition 2.11. *Let $f : X \rightarrow S$ be a morphism of smooth k -schemes. If $(*)$ satisfies base change, there exists a canonical flat connection ∇ on $H_{\mathrm{dR}}^i(X/S)$.*

The next section is devoted to the proof of this base-change formula.

3. 6-FUNCTOR FORMALISMS

Because of the non-algebraic nature of S_{dR} , establishing the above base change is slightly delicate; since S_{dR} is not a scheme, we cannot apply traditional proper or smooth base change for quasi-coherent sheaves on schemes. To this end, we proceed by setting up some abstract machinery: we appeal to the six functor formalism for quasi-coherent sheaves on X_{dR} as found in [Rib24] and [Rod25]. In this section, assume that all functors are appropriately derived.

3.1. Abstract 6-functor formalisms. We recall the definition of a 6-functor formalism. Rather than strive for an exhaustive treatment, we will introduce only the minimal amount of theory to state the desired result. For details see [Sch22].

Definition 3.1. A *geometric setup* is a pair $(\mathcal{C}, E)$ where $\mathcal{C}$ is an ∞ -category with finite limits and E is a collection of morphisms in $\mathcal{C}$ containing all isomorphisms and stable under pullback, composition, and taking diagonals.

Remark 3.2. E is the class of $!$ -able morphisms, or morphisms which admit an $f_!$.

Definition 3.3. Given a geometric setup $(\mathcal{C}, E)$, the category of correspondences $\text{Corr}(\mathcal{C}, E)$ is defined to be the category whose objects are the objects of $\mathcal{C}$ and a morphism from X to Y is a diagram

$$\begin{array}{ccc} & Z & \\ \swarrow & & \searrow \\ X & & Y \end{array}$$

in $\mathcal{C}$ where the arrow on the right $Z \rightarrow Y$ lies in E . A composition of morphisms is

$$\begin{array}{ccccc} & & Z \times_Y S & & \\ & \swarrow & & \searrow & \\ & Z & & S & \\ \swarrow & & \searrow & & \swarrow \\ X & & Y & & T \end{array}$$

where the left arrow $Z \times_Y S \rightarrow T$ lies in E by assumption. There is a monoidal structure on $\text{Corr}(\mathcal{C}, E)$ given by the product in $\mathcal{C}$.

Definition 3.4. Given a geometric setup $(\mathcal{C}, E)$, a 3-functor formalism is a lax symmetric monoidal functor $\mathcal{D} : \text{Corr}(\mathcal{C}, E) \rightarrow \text{Cat}_\infty$.

Given an arbitrary morphism $f : X \rightarrow Y$ in $\mathcal{C}$, the image of the correspondence

$$\begin{array}{ccc} & X & \\ f \swarrow & & \searrow \\ Y & & X \end{array}$$

is defined to be $f^* : \mathcal{D}(Y) \rightarrow \mathcal{D}(X)$. If $f : X \rightarrow Y$ lies in E , the image of

$$\begin{array}{ccc} & X & \\ \parallel \swarrow & & \searrow f \\ X & & Y \end{array}$$

is defined to be $f_! : \mathcal{D}(X) \rightarrow \mathcal{D}(Y)$.

Definition 3.5. A 6-functor formalism is a 3-functor formalism

$$\mathcal{D} : \text{Corr}(\mathcal{C}, E) \rightarrow \text{Cat}_\infty$$

such that the following holds:

- (a) the functors f^* and $f_!$ both admit right adjoints (defined to be f_* and $f^!$),
- (b) for each object of $\mathcal{C}$, the symmetric monoidal ∞ -category $\mathcal{D}(X)$ is closed.

A formal consequence of this is the following: given any Cartesian diagram

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ f' \downarrow & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

in $\mathcal{C}$ where f is $!$ -able, there is a specified isomorphism of functors

$$g^* \circ f_! \rightarrow f'_! \circ (g')^*$$

[Man22]. This is a salient feature of 6-functor formalisms and the motivation for our consideration of them.

Definition 3.6. Let $f : X \rightarrow Y$ be $!$ -able and $\omega_f := f^!(1_Y)$ be the dualizing object. The morphism f is said to be *cohomologically smooth* provided that

- (a) The natural transformation $\omega_f \otimes f^*(-) \rightarrow f^!(-)$ is an equivalence,
- (b) ω_f is $\otimes$ -invertible in $\mathcal{D}(X)$,
- (c) For any morphism $g : Y' \rightarrow Y$ with base change $f' : X' \rightarrow Y'$ of f , the above properties also hold for f' . Moreover, if $g' : X' \rightarrow X$ denotes the base change of g , we also require the natural map

$$(g')^* \omega_f \rightarrow \omega_{f'}$$

to be an isomorphism.

For more details see [Sch22] §5. Cohomological smoothness is important because it yields abstract notion of smooth base change ([Zav23], lemma 2.3.9).

Theorem 3.7 (Cohomologically smooth base change). *Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ f' \downarrow & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

be a Cartesian square in $\mathcal{C}$. Then the natural morphism

$$g^* \circ f_* \rightarrow (f')_* \circ (g')^*$$

is an isomorphism if g is cohomologically smooth.

As a final point, we mention briefly a particular type of cohomologically smooth morphism called cohomologically étale morphisms. Intuitively, cohomologically étale morphisms are cohomologically smooth with trivial dualizing object, i.e. satisfying $f^* \cong f^!$. We do not include a precise definition since cohomological smoothness is a sufficient condition for our needs (see [Zav23] for a precise definition).

3.2. Quasi-coherent sheaves on algebraic stacks. We outline a construction for a 6-functor formalism for the derived category of quasi-coherent sheaves on algebraic stacks. This relies on extension theorems for 6-functor formalism which we will treat as a black box. This is done in full detail in [Rod25] and [Rib24].

As before, let Aff_k be the category of affine k -schemes.

Theorem 3.8. *The functor $D_{\text{qc}} : \text{Aff}_k^{\text{op}} \rightarrow \text{Cat}_{\infty}$ which maps X to its derived ∞ -category of quasi-coherent sheaves $D_{\text{qc}}(X)$ extends to a 6-functor formalism on the geometric setup $(\text{Aff}_k, \text{all})$ consisting of all morphisms.*

This is proved in [Sch22] §8.3. To extend this to a 6 functor formalism on algebraic stacks, we appeal to results of [HM24]. This uses the D_{qc} -topology, though we will not elaborate on this since we can restrict to the coarser fppf topology.

Theorem 3.9. *The six functor formalism on the geometric setup $(\text{Aff}_k, \text{all})$ extends uniquely to a six functor formalism $(\text{AlgStk}_{D_{\text{qc}}}, E)$ such that*

- (a) *Let $f : Y \rightarrow X$ be a map whose pullback to every object in Aff_k lies in E , then f lies in E .*
- (b) *Let $f : Y \rightarrow X$ be a map which is $!$ -locally on the source or target in E ; then f lies in E .*
- (c) *Every map $f : Y \rightarrow X$ in E with $X \in \text{Sch}$ is $!$ -locally on the source in E .*

We emphasize that we do not have explicit knowledge of the class E . To check that a particular morphism of algebraic stacks is $!$ -able, we need to verify one of the above conditions. For a detailed account, see [Sch22] *Appendix to Lecture IV: Passage to stacks* and [Rod25], Corollary 6.19.

Remark 3.10. The above result uses the D_{qc} -topology on algebraic stacks. However this does not pose a substantial problem; the D_{qc} -topology is *finer* than the fppf topology, and [Rib24] shows that there is a unique six functor formalism on fppf algebraic stacks whose $!$ -able morphisms are precisely the morphisms whose sheafifications lie in the E above (proposition 1.3.23). That is, there is a compatible 6-functor formalism on the geometric setup $(\text{AlgStk}_{\text{fppf}}, E')$. In the sequel, we restrict to this formalism.

We need the following technical lemmas to proceed. Let X and Y be schemes.

Lemma 3.11 ([Sch22], Proposition 8.14). *If $f : X \rightarrow Y$ is smooth and proper, then f is cohomologically smooth.*

Lemma 3.12 ([Rib24], Proposition 1.3.25). *Let $Z \subset X$ be a closed subscheme, where the ideal sheaf is locally finitely generated. Then the morphism*

$$\widehat{X}_Z \rightarrow X$$

is cohomologically étale.

With these, we can prove the following central lemma.

Proposition 3.13. *If X is smooth over k , the natural morphism $X \rightarrow X_{\mathrm{dR}}$ is $!$ -able and cohomologically smooth.*

We remark that the proof of this result is essentially ([Sch22], proposition 8.23), but Scholze works in the IndCoh-formalism.

Proof. Cohomological smoothness can be verified locally on the source and target. Since this is local on the source, we can restrict to an affine U . Moreover, by smoothness, we can assume U is of finite presentation over k . By taking a compactification, we can restrict to a proper U . Let $Y \rightarrow U_{\mathrm{dR}}$ be a morphism with Y locally of finite presentation over k . Then the following diagram is Cartesian,

$$\begin{array}{ccc} (\widehat{U \times_k Y})_Z & \longrightarrow & Y \\ \downarrow & & \downarrow \\ U & \longrightarrow & U_{\mathrm{dR}} \end{array}$$

where $Z \subset U \times_k Y$ is the graph of $Y_{\mathrm{red}} \rightarrow U$, and Z is closed because U is separated and $Y_{\mathrm{red}} \rightarrow Y$ is a closed immersion. Then $(\widehat{U \times_k Y})_Z \rightarrow Y$ is the composite of $(\widehat{U \times_k Y})_Z \rightarrow U \times_k Y$ and $U \times_k Y \rightarrow Y$. Since the ideal sheaf of Z is finitely generated, the first morphism is cohomologically smooth by 3.12. The second is a base change of $U \rightarrow \mathrm{Spec} k$ and is thus cohomologically smooth by 3.11. $\square$

Proposition 3.14. *Let $f : X \rightarrow Y$ be a morphism between smooth, finite type k -schemes. Then $f_{\mathrm{dR}} : X_{\mathrm{dR}} \rightarrow Y_{\mathrm{dR}}$ is $!$ -able in the 6-functor formalism for quasi-coherent sheaves.*

Proof. Since the class of $!$ -able morphisms is stable under composition, it suffices to verify this after precomposing with the cohomologically smooth map $X \rightarrow X_{\mathrm{dR}}$. Then $X \rightarrow Y_{\mathrm{dR}}$ is the composite of f and $Y \rightarrow Y_{\mathrm{dR}}$, which are both $!$ -able. $\square$

3.3. The Gauss-Manin connection. Let $f : X \rightarrow S$ be a morphism of smooth, finite type k -schemes and consider the diagram

$$\begin{array}{ccc} (X/S)_{\mathrm{dR}} & \xrightarrow{\tilde{\pi}} & X_{\mathrm{dR}} \\ \tilde{f} \downarrow & & \downarrow f_{\mathrm{dR}} \\ S & \xrightarrow{\pi} & S_{\mathrm{dR}} \end{array} \quad (*)$$

Then, 3.7 and 3.13 imply the following:

Proposition 3.15. *The natural transformation*

$$\pi^* R^i f_{\mathrm{dR}*} \rightarrow R^i \tilde{f}_* \tilde{\pi}^*$$

is an equivalence.

We emphasize that the smoothness of f is unnecessary to establish base change for the relative de Rham square; since π is cohomologically smooth we need not assume any conditions on f_{dR} (other than that it's !-able).

Corollary 3.16. *There is a canonical flat connection on the relative de Rham cohomology $H_{\mathrm{dR}}^i(X/S)$.*

This $\mathcal{D}_S$ -module is naturally isomorphic to the i -th higher direct image of the structure sheaf of X_{dR} along f_{dR} ,

$$(H_{\mathrm{dR}}^i(X/S), \nabla) \cong R^i f_{\mathrm{dR}*} \mathcal{O}_{X_{\mathrm{dR}}}$$

Remark 3.17. We can also state the following slightly more general version: let $(\mathcal{F}, \nabla)$ be a left $\mathcal{D}_X$ -module, and let $\mathrm{dR}_{X/S}(\mathcal{F}) = \mathcal{F} \otimes_{\mathcal{O}_S} \Omega_{X/S}^\bullet$ be the relative de Rham complex. By the relative version of 2.9, we have an isomorphism

$$\mathbb{R}^i f_*(\mathrm{dR}_{X/S}(\mathcal{F})) \cong \pi^* R^i f_{\mathrm{dR}*} \mathcal{F}'$$

where $\mathcal{F}' = \Phi(\mathcal{F}, \nabla)$ is the quasi-coherent sheaf on X_{dR} corresponding to $(\mathcal{F}, \nabla)$. Consequently the right-hand side has a natural flat connection, which generalizes the Gauss-Manin connection to “de Rham cohomology” with coefficients in $(\mathcal{F}, \nabla)$.

Remark 3.18. This also provides a compact proof of the spectral sequence

$$\mathbb{H}^i(S, H_{\mathrm{dR}}^j(X/S) \otimes_{\mathcal{O}_S} \Omega_S^\bullet) \implies H_{\mathrm{dR}}^{i+j}(X/k)$$

described by [KO68]. This is the Leray spectral sequence for $f_{\mathrm{dR}} : X_{\mathrm{dR}} \rightarrow S_{\mathrm{dR}}$

$$H^i(S_{\mathrm{dR}}, R^j f_{\mathrm{dR}*} \mathcal{O}_{X_{\mathrm{dR}}}) \implies H^{i+j}(X_{\mathrm{dR}}, \mathcal{O}_{X_{\mathrm{dR}}})$$

by 2.9.

4. APPENDIX: GENERALITIES ON COHOMOLOGY THEORIES

The Gauss-Manin connection on $H_{\mathrm{dR}}^i(X/S)$ is a particular case of a general construction. Let k be a field of arbitrary characteristic, and τ a Grothendieck topology on Aff_k . We make the following nonstandard definition.

Definition 4.1. A $\heartsuit$ -cohomology theory is a cohomology theory for some class $\mathcal{S}$ of k -schemes with an associated 6-functor formalism $\mathrm{D}_{\heartsuit}$ satisfying: For any $X \in \mathcal{S}$, there is a functorial assignment $X \rightsquigarrow X_{\heartsuit}$, where the latter is a τ algebraic stack along with a morphism $X \rightarrow X_{\heartsuit}$. This data is required to induce

(a) an isomorphism of cohomology,

$$H_{\heartsuit}^i(X/k) \cong R^i\Gamma(X_{\heartsuit}, \mathcal{O})$$

and likewise for relative cohomology with $(X/S)_{\heartsuit} := X_{\heartsuit} \times_{S_{\heartsuit}} S$,

(b) and an equivalence of 6-functor formalisms,

$$D_{\heartsuit}(X) \cong D_{\text{qc}}(X_{\heartsuit})$$

In this context, we often call $D_{\heartsuit}$ the coefficients for the cohomology theory $H_{\heartsuit}$.

Remark 4.2. de Rham, Dolbeault, crystalline, and prismatic cohomology are all examples of $\heartsuit$ -cohomology theories for some suitable class of k -schemes, with associated coefficients $\mathcal{D}$ -modules, Higgs sheaves, crystals, and F -gauges, respectively. The assignment $X \rightsquigarrow X_{\heartsuit}$ is sometimes called *transmutation* [Bha22].

Let $f : X \rightarrow S$ be a morphism in $\mathcal{S}$ and consider the induced $f_{\heartsuit} : X_{\heartsuit} \rightarrow S_{\heartsuit}$.

Proposition 4.3. *Suppose $X \rightarrow X_{\heartsuit}$ is cohomologically smooth for D_{qc} . Then the relative cohomology $H_{\heartsuit}^i(X/S)$ has a canonical $\heartsuit$ -coefficient structure via $R^i(f_{\heartsuit})_* \mathcal{O}_{X_{\heartsuit}}$.*

To summarize, we always have a natural additional structure on relative cohomology induced from transmutation $X \rightsquigarrow X_{\heartsuit}$.

4.1. The Kodaira-Spencer map. Let k be a field of characteristic 0 and X a smooth k -scheme. Recall that the *algebraic Dolbeault cohomology* is defined to be

$$H_{\text{Dol}}^i(X/k) := \bigoplus_{p+q=i} R^p f_* \Omega_{X/k}^q$$

Dolbeault cohomology is a $\heartsuit$ -cohomology theory in the sense of 4.1 [Sim99]: there exists a fppf algebraic stack X_{Dol} and a cover $X \rightarrow X_{\text{Dol}}$ whose $\mathcal{O}$ -coherent cohomology recovers H_{Dol} . The category of quasi-coherent sheaves on X_{Dol} is equivalent to the category of quasi-coherent *Higgs* sheaves on X . The stack X_{Dol} has a simplicial presentation as $\widehat{T^{\bullet}X}$, where $T^n X$ is the n -fold product $TX \times_X \cdots \times_X TX$ and the formal completion is along the zero section $X \rightarrow T^n X$. By the proof of 3.13,

Corollary 4.4. *If X is smooth over k , $X \rightarrow X_{\text{Dol}}$ is cohomologically smooth. Consequently, there is a natural Higgs field ϑ on $H_{\text{Dol}}^i(X/S)$ and*

$$(H_{\text{Dol}}^i(X/S), \vartheta) \cong R^i f_{\text{Dol}*} \mathcal{O}_{X_{\text{Dol}}}$$

If we assume $f : X \rightarrow S$ is a smooth morphism, this is precisely the Higgs structure induced via the Kodaira-Spencer map $\kappa : \Omega_S^{\vee} \rightarrow R^1 f_* \Omega_{X/S}^{\vee}$. Cup product with the associated section in $R^1 f_* \Omega_{X/S}^{\vee} \otimes \Omega_S$ gives an $\mathcal{O}_S$ -linear map

$$R^p f_* \Omega_{X/S}^q \rightarrow R^{p+1} f_* \Omega_{X/S}^{q-1} \otimes_{\mathcal{O}_S} \Omega_S$$

This yields an $\mathcal{O}_S$ -linear map $\vartheta : H_{\text{Dol}}^i(X/S) \rightarrow H_{\text{Dol}}^i(X/S) \otimes_{\mathcal{O}_S} \Omega_S$ with $\vartheta \wedge \vartheta = 0$.

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